Network-Based Diversification of Stock Portfolios

Network Science: Project Presentation

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Presentation Overview

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Financial markets display complex dynamics:

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- There aren't many techniques that can capture such dynamics.

Introduction

Motivation

Financial markets display complex dynamics:

- There aren't many techniques that can capture such dynamics.
- Network science is a promising framework to capture these intricasies.

Traditional approaches to investing:

Introduction

Motivation

Traditional approaches to investing:

- Maximize diversification.

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Using network science to automatically build a portfolio:

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Using network science to automatically build a portfolio:

- Community detection algorithms identify distinct market segments.

Introduction

Motivation

Traditional approaches to investing:

- Maximize diversification.
- Minimize volatility.

Using network science to automatically build a portfolio:

- Community detection algorithms identify distinct market segments.
- Network metrics identify the most desirable assets within each segment.

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- Correlation measures.
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Their approach was then tested on the S&P 500 index.

Building on this work, we:

Introduction

Contributions

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- Explore different community detection algorithms.

Building on this work, we:

- Explore different correlation measures.
- Explore different community detection algorithms.
- Analyze our approach on a wider subset of the global financial markets.

Automated Portfolio Construction

Overview

Three steps:

Automated Portfolio Construction

Overview

Three steps:

- Compute relations between the daily returns of assets.

Automated Portfolio Construction

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- Identify market segments based on these relations.

Automated Portfolio Construction

Overview

Three steps:

- Compute relations between the daily returns of assets.
- Identify market segments based on these relations.
- Build a portfolio by selecting assets from each segment.

Automated Portfolio Construction

Asset Relations

Computing asset relations:

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Asset Relations

Computing asset relations:

- Spearman Rank Correlation Coefficient (SCC).

Automated Portfolio Construction

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- Spearman Rank Correlation Coefficient (SCC).
- Pearson Correlation Coefficient (PCC).

Automated Portfolio Construction

Asset Relations

Computing asset relations:

- Spearman Rank Correlation Coefficient (SCC).
- Pearson Correlation Coefficient (PCC).
- Co-Occurrence measures.

Automated Portfolio Construction

Asset Relations

Four distinct Co-Occurrence measures:

Automated Portfolio Construction

Asset Relations

Four distinct Co-Occurrence measures:

- Co-Occurrence SCC using Louvain to identify the monthly communities.

Automated Portfolio Construction

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Asset Relations

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Automated Portfolio Construction

Asset Relations

Computing a Co-Occurrence measure:

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Asset Relations

Computing a Co-Occurrence measure:

- Compute the monthly SCC/PCC relations between assets.

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Asset Relations

Computing a Co-Occurrence measure:

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- Identify the market segments for each month using a community detection algorithm.

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Asset Relations

Computing a Co-Occurrence measure:

- Compute the monthly SCC/PCC relations between assets.
- Identify the market segments for each month using a community detection algorithm.
- For each pair of assets, compute the frequency with which they belong to the same community.

In short, we have 6 possibile measures of asset relations:

- SCC.
- PCC.
- Four Co-Occurrence measures.

Automated Portfolio Construction

Network Representation

We now have, for each pair of assets, a number reflecting how they are related.

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Network Representation

We now have, for each pair of assets, a number reflecting how they are related.

Goal: Use these values to determine groups of assets that are closely related.

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Network Representation

Build a fully connected graph:

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Network Representation

Build a fully connected graph:

- One node for each available asset.

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Network Representation

Build a fully connected graph:

- One node for each available asset.
- Use the relation measures between assets as edge weigths.

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Network Representation

Problem: Most community detection algorithms are designed for sparse networks.

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- Our fully connected graph is not suitable for this.

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Solution: Compute the *Minimum Spanning Tree* of our graph.

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Network Representation

Problem: Most community detection algorithms are designed for sparse networks.

- Our fully connected graph is not suitable for this.

Solution: Compute the *Minimum Spanning Tree* of our graph.

- Yields a graph amenable to community detection.

Automated Portfolio Construction

Community Detection

Identifying market segments:

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Community Detection

Identifying market segments:

- Louvain Method.

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Community Detection

Identifying market segments:

- Louvain Method.
- Girvan-Newman Algorithm.

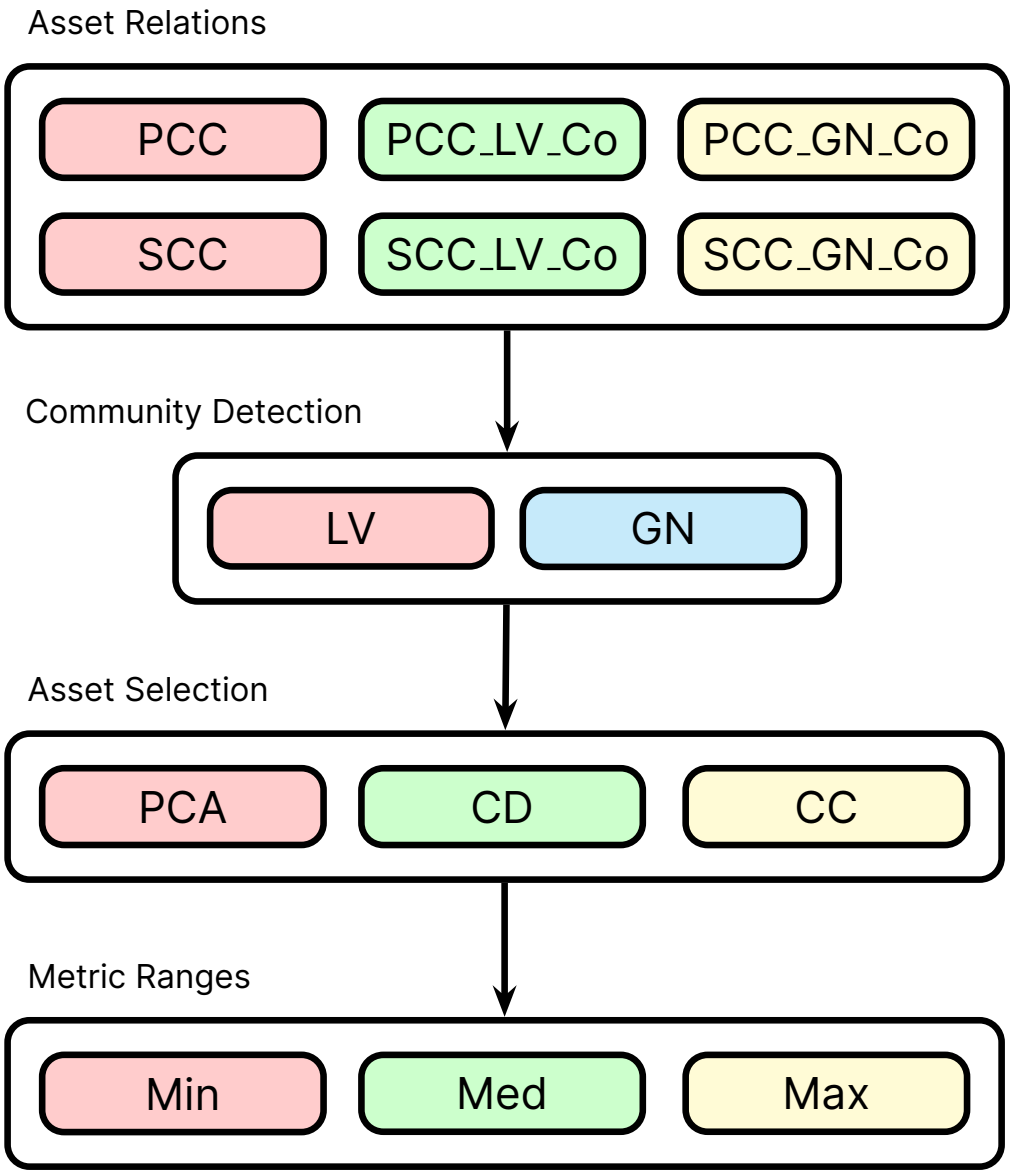
Automated Portfolio Construction

Asset Selection

TODO

Automated Portfolio Construction

Summary



Results

Performance

TODO

Results

Robustness

TODO

Bibliography

[1] Dimitar Kitanovski et al. “Network-based diversification of stock and cryptocurrency portfolios”. In: *Applied Network Science* 10.20 (2025), pp. 1–46. DOI: 10.1007/s41109-025-00708-9.